

Yuchen Wang

Agentic post-training | ML systems

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Education

University of Illinois Urbana-Champaign

M.S. in Computer Science

2025 – 2027 (expected)

Champaign, IL

Peking University

B.S. in Intelligent Science and Technology (AI); Zhi Class (Elite Program), Top 10%

2021 – 2025

Beijing, China

Research Experience

Alibaba U.S. – Accio Team

May 2026 – Present

Research Intern, Agentic LLM Post-Training

Sunnyvale, CA

- Core contributor to **Occamy-1.0** [[arXiv](#) · [Website](#)], a 35B-A3B agent model. Built execution-grounded data and post-training infrastructure for long-horizon tool use; the team's SFT corpus contains **15K trajectories and 403M tokens**.
- Developed verifier-gated pipelines for OpenClaw, Hermes, and Accio Work: token-exact replay, state recovery, and compaction-aware segments preserving tokens, log-probs, and loss masks for SFT/RL.
- The released model improves Claw-Eval average from **69.5 to 82.2** and AutomationBench strict pass from **7.5% to 27.6%**, reaching the Pareto frontier in the report's four-benchmark cost comparison.

Freedo Technology

May – Aug 2025

Research Intern, 3D Vision and Generative AI

Beijing, China

- Built depth/normal-conditioned ControlNet for 3D building reconstruction from noisy point clouds, preserving structural edges and planar surfaces; improved geometric fidelity from **32.7% to 85.3%**.

Open-Source Systems Engineering

- **Inference and distributed performance.** [vLLM](#): in-place BF16 MoE conversion halved TP2 conversion peak allocation (**7.88 to 3.94 GiB**). [NeMo RL](#): deferred teacher top-k payloads, achieving **4.44–5.31× speedup** in controlled Ray transfer measurements.
- **Numerics and long-context training.** Fixed [FlashInfer FP8 KV calibration](#) and [small-norm Muon normalization](#); fused [GatedDeltaNet Q/K normalization](#) and added selective Mamba recompute in Megatron-LM.
- **Agent runtime reliability.** Repaired [schema-reference argument coercion](#) and [SSH teardown/reconnect](#) in Hermes; co-authored [pickle-safe cross-process HTTP failures](#) in NeMo Gym.

Selected Research Projects

CineFlow: Dependency-Driven Parallel Video Generation [[Paper](#)]

Oct 2025 – May 2026

Advisor: Prof. Fan Lai, UIUC

- Built semantic DAGs, runtime trajectory fusion, online edge pruning, and critical-path scheduling for video diffusion without retraining; analyzed dependency structure in 1K+ MovieGenBench prompts.
- Across Wan2.2-5B, CogVideoX-5B, and HunyuanVideo on 8×H100: **1.7–5.5× end-to-end speedup**, 1.30–2.02× lower P90 latency, and **5.4–17.3% higher VBench overall**.

Dynamic Prefill Optimization via Adaptive Online Packing [[Report](#)]

Aug 2025 – Jan 2026

University of Illinois Urbana-Champaign

- Built AIMD batching control using live p95 TTFT feedback and greedy/DP length-aware packing; reduced TTFT by **up to 20%** on production-style DynamoLLM traces.

FlashAttention-style CUDA Kernels for GPT-2 [[Report](#)]

Aug – Dec 2025

University of Illinois Urbana-Champaign

- Implemented tiled attention, online softmax, and kernel fusion: arithmetic intensity **15 to 157 FLOP/byte**, roughly **10× lower HBM traffic**, and up to 9% end-to-end speedup.

Reinforcement Learning for Legal Reasoning [[Thesis](#)]

Jan – Mar 2025

Advisor: Prof. Yansong Feng, Peking University

- Built a Zero-RL → distilled-CoT SFT → GRPO pipeline; reached **57.6% accuracy**, outperforming SFT-only and RL-only baselines.

Technical Skills

Languages: Python, C++, CUDA. **Systems:** PyTorch, Ray, FSDP, Megatron-LM, NeMo RL, vLLM, SGLang, Nsight.
Methods: RL, distributed training, diffusion transformers, ControlNet, 3D vision and point clouds.